



How to Play with a Child Holding Mythic Weapons

Moxian Qian · Helmholtz Institute Mainz (HIM)

The Master's Four Manuals

The weapon can search.
The manuals decide what may
be believed and written.



The Quest Scroll

The quest begins
with a clear mission.

- 
- 1 verbal clue
 - 2 target paper
 - 3 expand and read
 - 4 evidence and boundaries



verbal clue



candidate pool



original papers



bounded claims

Responsible
Research,
Responsible
Power

The Quest Scroll



verbal clue

Moxian Qian's
recent stochastic
path sampler paper



target paper

Let $\mathcal{P}$ be the paper
set and $p^* \in \mathcal{P}$ the
target paper.

Search aims to build
a covering subset
 $S \subset \mathcal{P}$ around p^* .



expand and read

Expand by backward
citations, forward
citations, author links,
and method-family
neighbors; then read
selected papers to
improve coverage.



evidence and
boundaries

Original source: PDF, DOI,
title, authors, and exact
figure/page screenshot.

Boundary: what the source
supports — and does not
support.

Open question: a plausible
next claim not yet
supported.

Target Lock

Before entering the literature maze, the identity of the target paper must be locked.

C0
search result

C1
identity verified

C2
working page / PDF

C3
original context

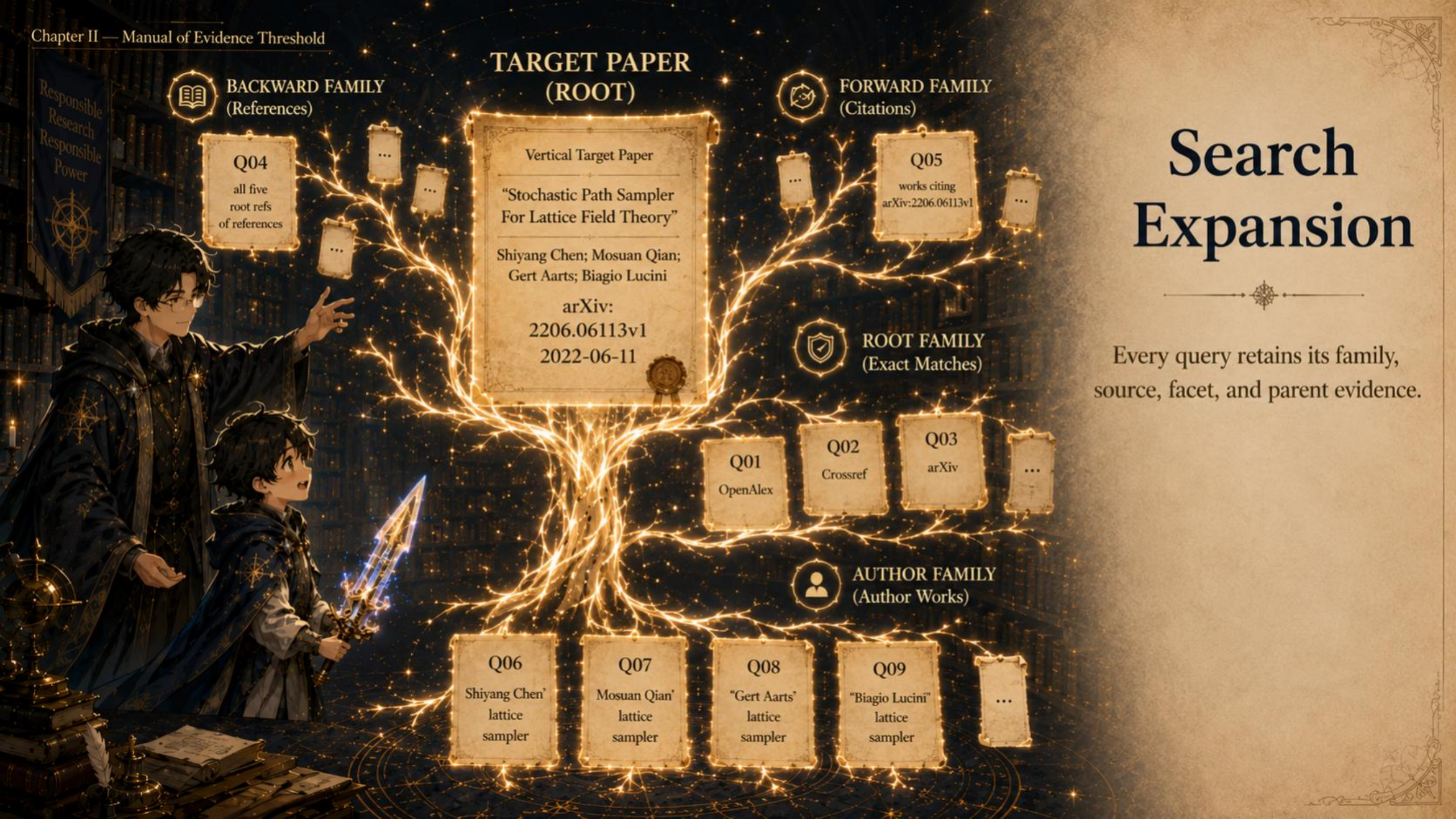
C4
claim with exact locator

Target Lock

Verified Target Paper

Field	Verified result
Title	Stochastic Path Sampler for Lattice Field Theory
Authors	Shiyang Chen; Moxian Qian; Gert Aarts; Biagio Lucini; Kai Zhou
arXiv ID	2606.13790v1
Date	2026-06-11
Primary PDF	arXiv page and downloaded PDF

Identity must be locked before citation and method lineages are expanded.



BACKWARD FAMILY (References)

Q04

all five
root refs
of references



TARGET PAPER (ROOT)

Vertical Target Paper

“Stochastic Path Sampler
For Lattice Field Theory”

Shiyang Chen; Mosuan Qian;
Gert Aarts; Biagio Lucini

arXiv:
2206.06113v1
2022-06-11



FORWARD FAMILY (Citations)

Q05

works citing
arXiv:2206.06113v1



ROOT FAMILY (Exact Matches)

Q01

OpenAlex

Q02

Crossref

Q03

arXiv



AUTHOR FAMILY (Author Works)

Q06

Shiyang Chen'
lattice
sampler

Q07

Mosuan Qian'
lattice
sampler

Q08

“Gert Aarts”
lattice
sampler

Q09

“Biagio Lucini”
lattice
sampler



Search Expansion

Every query retains its family,
source, facet, and parent evidence.

Candidate Screening

The final pool has 593 candidates.
Screening prioritizes reading;
a candidate is not yet evidence.



Candidate is not yet evidence.

Reading-Priority Ledger

arXiv ID	Title
arxiv:2606.13790	Stochastic Path Sampler For Lattice Field Theory
arxiv:2502.05504	Physics-Conditioned Diffusion Models for Lattice Gauge Theory
arxiv:2409.15937	Numerical determination of the width and shape of the effective string using Stochastic Normalizing Flows
arxiv:2412.19109	Stochastic normalizing flows for Effective String Theory
arxiv:2210.03139	Stochastic normalizing flows for lattice field theory

Reading Score	Decision
121	include
20	include
19	include
18	include
18	include

Overall Result
593
candidates

Dijkstra Ranking

The route decides where to look first.

Edge weight =
reading cost
(distance)



Goal: reach older, foundational sources at minimum cost.

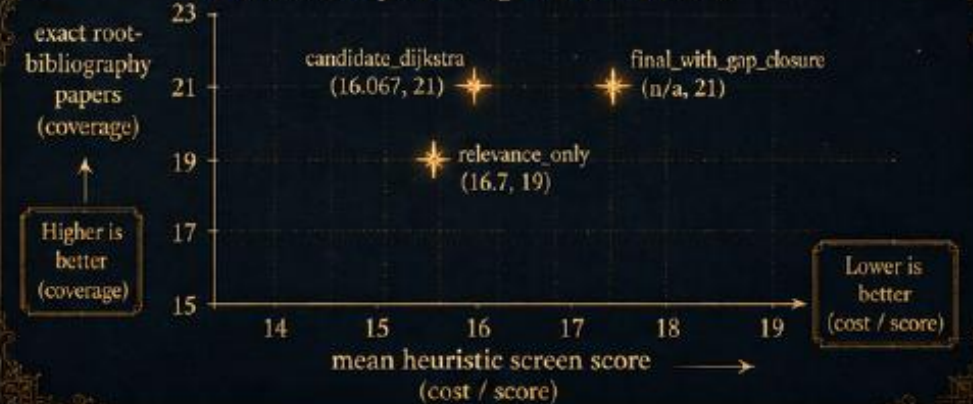
Dijkstra always expands the lowest-cost frontier first.

Navigation Outcomes (30-Paper Budget)

metric	relevance_only	candidate_dijkstra	final_with_gap_closure	interpretation
selected papers	30	30	37	fixed reading budget for the first two arms
declared facets covered	7	7	10	Dijkstra did not increase coarse facet count
method groups covered	8	8	8	coverage equality does not imply identical papers
exact root-bibliography papers	19	21	21	candidate Dijkstra retained two more root ancestors
mean heuristic screen score	16.7	16.067	not comparable	graph navigation traded a little keyword score for paths
selection overlap	20	20	35	20/30 papers overlap; ten are replaced

With the same 30-paper budget, Dijkstra retained two additional direct references of the target paper; coarse facets and method groups did not increase.

Outcome Map: Coverage vs. Cost (Score)



The route decides where to look first.

The Original-Source Library

A paper enters the report only
after its original text is opened.

Each source retains identity, relation, and primary URL.

Real Result

37

full texts read



10

facets



185

evidence entries



199

checked
direct-citation
relations

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Source Table

ID	arXiv ID	Title
P001	2606.13790	Stochastic Path Sampler For Lattice Field Theory
P002	2111.15141	Path Integral Sampler: a stochastic control approach for sampling
P003	2302.13834	Denoising Diffusion Samplers
P004	2307.01050	Transport meets Variational Inference: Controlled Monte Carlo Diffusions
P005	2410.02711	NETS: A Non-Equilibrium Transport Sampler
P006	2210.03139	Stochastic normalizing flows for lattice field theory
P007	2201.08862	Stochastic normalizing flows as non-equilibrium transformations



Identity

Who this source is



Relation

How it connects
to the report.



Primary URL

Where the original
text resides.

We read. We verify. We cite. Only then does it enter the report.

Problem and Method

Problem

- 1 Data-free sampling of an unnormalized lattice target near criticality, where ordinary chains develop long autocorrelation times.

Anchor: PDF p.1 abstract

- 2 Construct an efficient sampler for an unnormalized target by controlling a stochastic path rather than fitting a terminal density alone.

Anchor: PDF p.1 abstract

Every problem and method statement carries a PDF page or section anchor.



Method

- 1 Learn paired forward and auxiliary backward Langevin paths by minimizing a path-space variational free energy / entropy-production upper bound.

Anchor: PDF pp.4–8 Sec. 2

- 2 Formulate sampling as a Schrödinger-bridge / stochastic optimal-control problem and learn a neural control for a reference diffusion.

Anchor: PDF pp.4–8 method

We first identify the problem. Then we anchor the method to the original paper.

Results and Limits

No result travels without its limit.

Real Result

37

full texts read



10 facets



185 evidence entries



199 checked direct-citation relations



Correction / exactness

Independent path proposals are corrected with an extended-space independence Metropolis–Hastings step; Metropolis–Hastings enforces path likelihood ratios and removes residual bias.



Anchor: PDF pp.12–18 Sec. 3, Tables 1–4

Result

Corrected SPS observables agree with HMC in the tested 2D ϕ^4 systems and show shorter autocorrelation near the pseudocritical region.



Anchor: PDF pp.12–18 Sec. 3, Tables 1–4

Limitation

Uncorrected tails and multimodal regions deviate; finite network capacity, terminal diffusion and support coverage remain practical limits, and the reported timing is hardware/setup specific.



Anchor: PDF pp.12–13 Secs. 3.1–3.2; pp.19–21 discussion



Weighting / benchmarking

Girsanov path likelihood ratios provide importance weights; weighted estimates retain the path-measure correction.



Anchor: PDF pp.9–18 experiments

Result

The learned path sampler is competitive with the reported MCMC, SMC and normalizing-flow baselines on its benchmark suite.



Anchor: PDF pp.9–18 experiments

Limitation

Training overhead, time discretization and a suboptimal learned control can produce weight variance and residual bias if correction is omitted.



Anchor: PDF pp.19–20 discussion

Exactness, result, limitation, and their original anchors stay together.

Citation Lineage

Lineage shows relation, not identity.

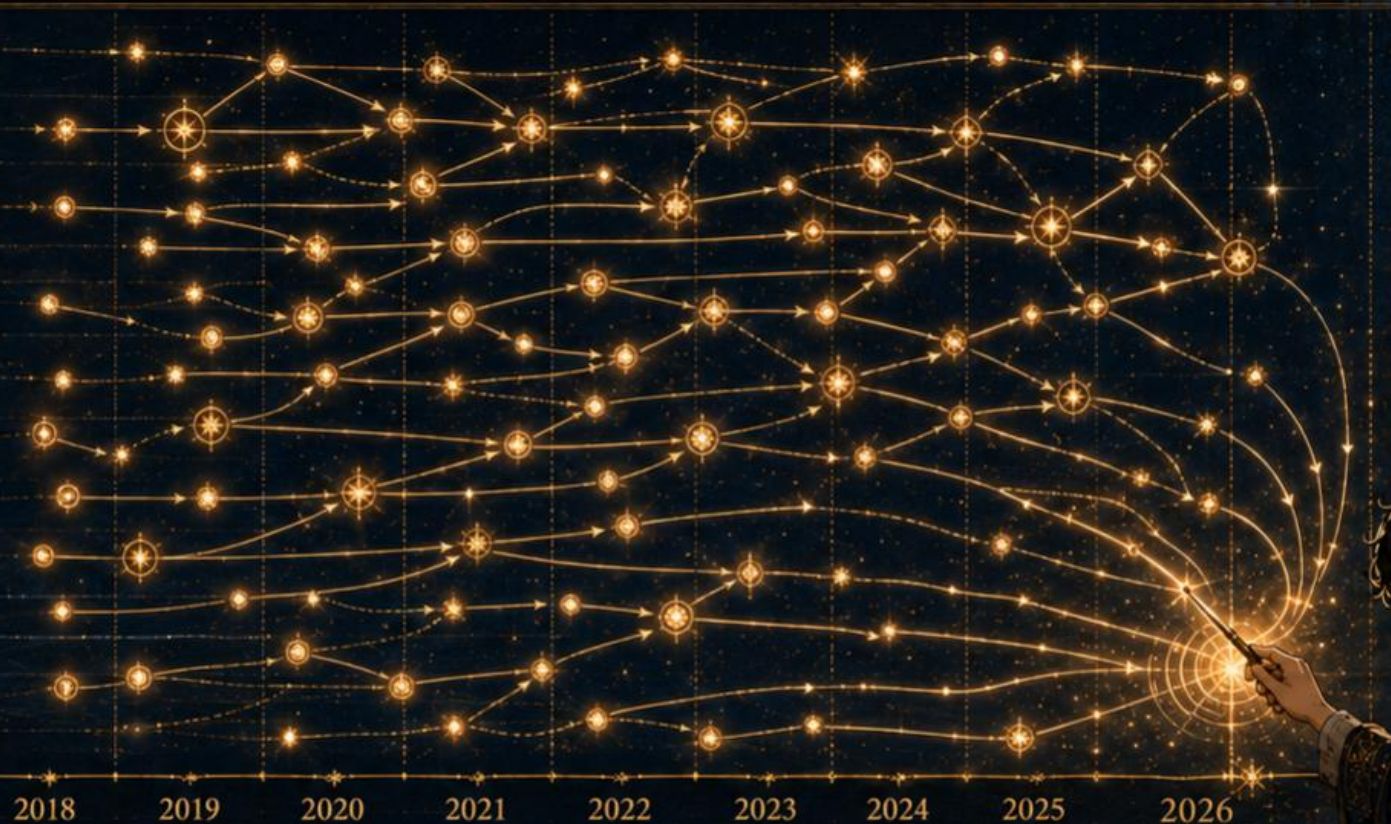
Method Family

-  Early generative baselines
-  Adjacent path generation
-  Path-space control
-  Normalizing flows
-  Stochastic flows
-  Diffusion
-  Autoregressive
-  Scaling and transfer
-  Scaling / multiscale
-  Diagnostics / theory
-  Extensions
-  Fermion and gauge applications
-  Flow observables
-  Normalizing-flow theory
-  Scaling and architecture
-  SPS

Legend

-  Checked paper
-  Highly influential (many citations)
-  Cross-family connection
-  Citation direction (cited → citing)

 Target paper
(2606.13790)



The horizontal axis is year, the vertical axis is method family, and arrows run from cited work to citing work.

Arrows run from cited work to citing work; only exact-identifier checked edges are shown.

Claims and Boundaries

The evidence determines the size of the claim.

claim_id · short sentence · boundary · status

Real Result

13
slides studied

10 facets

185 evidence entries

199 checked
direct-citation
relations

C01

Short sentence

SPS learns paired forward and auxiliary backward stochastic paths with a path-space variational objective.

Boundary

This is the SPS v1 formulation; do not replace it with terminal-density KL training.

Status

confirmed

C02

Short sentence

SPS follows learned independent path proposals with extended-space independence Metropolis–Hastings.

Boundary

Do not call uncorrected SPS samples exact.

Status

confirmed

C10

Short sentence

The defensible SPS follow-ups close correctness, scaling and physics-cost loops together.

Boundary

These are literature-grounded directions, not SPS results already achieved.

Status

confirmed

Claim, short sentence, boundary, and status must stay together.

Run Record

What the weapon actually did

Final Run

14

Slide

 37 full texts

 10 facets

 185 evidence entries

 199 checked direct-citation relations

Responsible
Research,
Responsible
Power

Run Record

Recorded item	Final run
 candidate graph	3,300 nodes / 8,187 edges; local Dijkstra took about 0.11 s.
 equal reading budget	relevance-only: 30 papers; Dijkstra: 30 papers; overlap: 20.
 selection effect	exact root-bibliography papers: 19 → 21.
 after gap closure	37 full texts; 10 facets; 185 evidence entries; 199 checked direct-citation relations.
 complete goal	1,012,083 tokens; 4,039 s; 19/19 validation gates PASS.
 not supported	Per-arm token/time was not measured; a Dijkstra path is navigation metadata, not scientific evidence.

The algorithm decides where to look first;
original sources and evidence gates decide what can be written.